Derivatives Pricing

A Computational & Theoretical Framework

©:Zac Kienzle[†]

[†] University of Queensland, Brisbane, Australia

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We start with a general model of a frictionless security market where investors are allowed to trade continuously up to some fixed finite planning horizon T. Uncertainty in the financial market is modelled by a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $\mathcal{F} = (\mathcal{F}_t)_{0 \le t \le T}$ satisfying the usual conditions of right-continuity and completeness. We assume that the σ-field $\mathcal{F}_0$ is trivial, i.e. for every $A \in \mathcal{F}_0$ either $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$, and that $\mathcal{F}_T = \mathcal{F}$.

There are $d+1$ primary traded assets, whose price processes are given by stochastic processes $S_0, \ldots, S_d$. We assume that $S = (S_0, \ldots, S_d)$ follows an adapted, right-continuous with left-limits (RCLL) and strictly positive semimartingale on $(\Omega, \mathcal{F}, \mathbb{P}, \mathcal{F})$. We assume that the processes $S_0, \ldots, S_d$ represent the prices of some traded assets (stocks, bonds, or options). We will occasionally use $\mathcal{M}(\mathbb{P})$ (or just $\mathcal{M}$ if reference to $\mathbb{P}$ is not needed) to denote a financial market model of the above type.

We have not emphasised so far that there was an implicit numéraire behind the prices $S_0, \ldots, S_d$; it is the numéraire relevant for domestic transactions at time t. The formal definition of a numéraire is very much as in the discrete setting.

Definition 11.1: numéraire

A numéraire is a price process $X(t)$ almost surely strictly positive for each $t \in [0, T]$.

We assume now that $S_0(t)$ is a non-dividend paying asset, which is (almost surely) strictly positive and we use S_0 as numéraire. *Historically* the money market account $B(t)$, given by $B(t) = \mathrm{e}^{rt}$ with a positive deterministic process $r(t)$ and $r(0) = 0$, was used as numéraire, and the reader may think of $S_0(t)$ as being $B(t)$.

Our principal task will be the pricing and hedging of contingent claims, which we model as $\mathcal{F}_T$-measurable random variables. This implies that the contingent claims specify a stochastic cash-flow at time T and that they may depend on the whole path of the underlying in $[0, T]$, because $\mathcal{F}_T$ contains all that information. We will often have to impose further integrability restrictions on the contingent claims under consideration. The fundamental concept in (arbitrage) pricing and hedging contingent claims is the interplay of self-financing replicating portfolios and risk-neutral probabilities. Although the current setting is on a much higher level of sophistication, the key ideas remain the same.

We call an $\mathbb{R}^{d+1}$-valued predictable, locally bounded process

$$\varphi(t) = \big(\varphi_0(t), \ldots, \varphi_d(t)\big), \quad t \in [0, T],$$

a trading strategy (or dynamic portfolio process). (The conditions ensure that the stochastic integral $\int_0^t \varphi(u)\, \mathrm{d}S(u)$ exists.) Here $\varphi_i(t)$ denotes the number of shares of asset i held in the portfolio at time t, to be determined on the basis of information available before time t; i.e. the investor selects his time t portfolio after observing the prices $S(t_-)$. The components $\varphi_i(t)$ may assume negative as well as positive values, reflecting the fact that we allow short sales and assume that the assets are perfectly divisible.

Definition 11.2: Trading Strategy, Value And Gains Processes

(i) The value of the portfolio φ at time t is given by the scalar product

$$V_\varphi(t) = \varphi(t) \cdot S(t) = \sum_{i=0}^{d} \varphi_i(t)\, S_i(t), \quad t \in [0, T]. \tag{11.1}$$

The process $V_\varphi(t)$ is called the value process, or wealth process, of the trading strategy φ.

(ii) The gains process $G_\varphi(t)$ is defined by

$$G_\varphi(t) = \int_0^t \varphi(u)\, \mathrm{d}S(u) = \sum_{i=0}^{d} \int_0^t \varphi_i(u)\, \mathrm{d}S_i(u). \tag{11.2}$$

(iii) A trading strategy φ is called self-financing if the wealth process $V_\varphi(t)$ satisfies

$$V_\varphi(t) = V_\varphi(0) + G_\varphi(t) \quad \text{for all } t \in [0, T]. \tag{11.3}$$

✍

(i) The financial implications of the above equations are that all changes in the wealth of the portfolio are due to capital gains, as opposed to withdrawals of cash or injections of new funds.

(ii) The definition of a trading strategy includes regularity assumptions in order to ensure the existence of stochastic integrals. In the current rather general setup these conditions are flexible enough to allow a discussion of the no-arbitrage condition; however, they prove too restrictive for the question of market completeness (see below).

Our objective is to be very flexible with regard to the chosen numéraire. The next results underline this.

Proposition 11.1: Numéraire Invariance Theorem

Self-financing portfolios remain self-financing after a numéraire change.

Proof. From a financial viewpoint this property is immediately clear. For a mathematical deduction we use the product rule to get

$$d\left(\frac{S_k(t)}{X(t)}\right) = S_k(t)\, d\left(\frac{1}{X(t)}\right) + \frac{1}{X(t)}\, dS_k(t) + d\langle S_k, 1/X\rangle(t), \tag{11.4}$$

where $\langle S_k, 1/X\rangle(t)$ denotes the quadratic covariation between the semimartingales S_k and $1/X$. In the same manner

$$d\left(\frac{V_\varphi(t)}{X(t)}\right) = V_\varphi(t)\, d\left(\frac{1}{X(t)}\right) + \frac{1}{X(t)}\, dV_\varphi(t) + d\langle V_\varphi, 1/X\rangle(t). \tag{11.5}$$

The self-financing condition implies that

$$\begin{aligned} d\left(\frac{V_\varphi(t)}{X(t)}\right) &= \sum_{k=1}^{d} \varphi_k(t) \left\{ S_k(t)\, d\left(\frac{1}{X(t)}\right) + \frac{1}{X(t)}\, dS_k(t) + d\langle S_k, 1/X\rangle(t) \right\} \\ &= \sum_{k=1}^{d} \varphi_k(t)\, d\left(\frac{S_k(t)}{X(t)}\right). \end{aligned} \tag{11.6}$$

Thus the portfolio expressed in the new numéraire remains self-financing. □

Turning back to the special numéraire $S_0(t)$ we consider the discounted price process

$$\widetilde{S}(t) \coloneqq \frac{S(t)}{S_0(t)} = \big(1, \widetilde{S}_1(t), \dots, \widetilde{S}_d(t)\big), \tag{11.7}$$

with $\widetilde{S}_i(t) \coloneqq S_i(t)/S_0(t),\ i = 1, 2, \dots, d$. Furthermore, the discounted wealth process $\widetilde{V}_\varphi(t)$ is given by

$$\widetilde{V}_\varphi(t) \coloneqq \frac{V_\varphi(t)}{S_0(t)} = \varphi_0(t) + \sum_{i=1}^{d} \varphi_i(t)\, \widetilde{S}_i(t). \tag{11.8}$$

and the discounted gains process $\widetilde{G}_\varphi(t)$ is

$$\widetilde{G}_\varphi(t) \coloneqq \sum_{i=1}^{d} \int_0^t \varphi_i(u)\, d\widetilde{S}_i(u). \tag{11.9}$$

Observe that $\widetilde{G}_\varphi(t)$ does not depend on the numéraire component φ_0.

It is convenient to reformulate the self-financing condition in terms of the discounted processes:

Proposition 11.2: Self-Financing In Discounted Terms

Let φ be a trading strategy. Then φ is self-financing if and only if

$$\widetilde{V}_\varphi(t) = \widetilde{V}_\varphi(0) + \widetilde{G}_\varphi(t). \tag{11.10}$$

Of course, $V_\varphi(t) \geq 0$ if and only if $\widetilde{V}_\varphi(t) \geq 0$.

✍ The above result shows that a self-financing strategy is completely determined by its initial value and the components $\varphi_1, \ldots, \varphi_d$. In other words, any set of predictable processes $\varphi_1, \ldots, \varphi_d$ such that the stochastic integrals $\int \varphi_i \mathrm{d}\widetilde{S}_i$ exist can be uniquely extended to a self-financing strategy φ with specified initial value $\widetilde{V}_\varphi(0) = v$ by setting

$$\varphi_0(t) = v + \sum_{i=1}^{d} \int_0^t \varphi_i(u)\, \mathrm{d}\widetilde{S}_i(u) - \sum_{i=1}^{d} \varphi_i(t)\, \widetilde{S}_i(t), \quad t \in [0, T].$$

11.1 Equivalent Martingale Measures

We develop a relative pricing theory for contingent claims. Again the underlying concept is the link between the no-arbitrage condition and certain probability measures.

Definition 11.3: Arbitrage Opportunity

A self-financing trading strategy φ is called an arbitrage opportunity if the wealth process V_φ satisfies the following set of conditions:

$$V_\varphi(0) = 0, \quad \mathbb{P}(V_\varphi(T) \geq 0) = 1, \quad \text{and} \quad \mathbb{P}(V_\varphi(T) > 0) > 0.$$

Arbitrage opportunities represent the limitless creation of wealth through risk-free profit and thus should not be present in a well-functioning market.

The main tool in investigating arbitrage opportunities is the concept of equivalent martingale measures:

Definition 11.4: Equivalent Martingale Measure

We say that a probability measure Q defined on $(\Omega, \mathcal{F})$ is a (*strong*) *equivalent martingale measure* if:

(i) Q is equivalent to $\mathbb{P}$,

(ii) the discounted price process $\tilde{S}$ is a Q-local martingale (martingale).

We denote the set of martingale measures by $\mathcal{P}$.

Lemma 11.1: Growth Rates

Assume $S_0(t) = B(t) = e^{r(t)}$, then $Q \sim \mathbb{P}$ is a martingale measure if and only if every asset price process S_i has price dynamics under Q of the form

$$dS_i(t) = r(t)S_i(t)dt + dM_i(t),$$

where M_i is a Q-local martingale.

In order to proceed we have to impose further restrictions on the set of trading strategies.

Definition 11.5: Tame Strategy

A self-financing trading strategy φ is called *tame* (relative to the numéraire S_0) if

$$\tilde{V}_\varphi(t) \geq 0 \quad \text{for all } t \in [0, T].$$

We use the notation Φ for the set of tame trading strategies.

We next analyse the value process under equivalent martingale measures for such strategies.

Proposition 11.3: Value Process Martingale

For $\varphi \in \Phi$, $\tilde{V}_\varphi(t)$ is a non-negative local martingale, and also a supermartingale, under each $Q \in \mathcal{P}$.

Proof. Since $\tilde{S}(t)$ is a local martingale and $\varphi \in \Phi$ is predictable and locally bounded, it follows that the stochastic integral

$$\tilde{G}_\varphi(t) = \int_0^t \varphi(u) d\tilde{S}(u)$$

is a local martingale. Now $\tilde{V}_\varphi(t) = \tilde{V}_\varphi(0) + \tilde{G}_\varphi(t)$, so $\tilde{V}_\varphi(t)$ is a local martingale, and non-negative since φ is tame. But non-negative local martingales are supermartingales. □

This observation is the key to our first central result:

Theorem 11.1: Market Model Arbitrage

Assume $\mathcal{P} \neq \emptyset$. Then the market model contains no arbitrage opportunities in Φ.

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Monte Carlo Methods

Simulation techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate simulated option price values, which are then compared against outcomes derived from analytical formulas.

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